DC 10–16 V

J101

Power Switch Connector

J116
JST

PSU_LNB

VIN
PW_SW1
PW_SW2
LNB_FLT
IMON
5V5_FLT
PSU_FAULT
LNB_SHDN
LNB_POL
VIN_Sense
LNBV_Sense
5V5_Sense
3V6_Sense

LNB_FLT
IMON
5V5_FLT
PSU_FAULT
LNB_SHDN
LNB_POL

VIN_Sense
LNBV_Sense
5V5_Sense
3V6_Sense

DIGITAL

File: PSU_LNB.kicad_sch

DET_OUT

ANALOG

LNB

J102
SMA

RX_LNB
File: RX_LNB.kicad_sch

LNB
RX_SDR
ATT_LE
ATT_CLK
ATT_DATA
RX_LNA_EN
RX_TRX
3V3_RF

Ext. RX

J109
SMA

TRX

J103
SMB

Input_&_IF_Mixer
File: Input_&_IF_Mixer.kicad_sch

TRX
3V3_RF
RX_TRX
ATT_LE
ATT_CLK
ATT_DATA
TX_EN
Vfwd
IF_TX
LO1

TX_mixer
File: TX_mixer.kicad_sch

IF_TX
RF_out
IF_LNA_EN
IF_LNA_EN
TX_AMP_EN
2nd_LO

Output_PA
File: Output_PA.kicad_sch

3V3_RF
RF_IN
TX_PA_EN
I2C_SCL
I2C_SDA
DET_OUT
PA_Out
Sample

2.4 GHz TX PA
J114
SMA

~-23 dB sample
J115
SMA

J106
SMB
1st LO
~400 MHz

J110
SMB
1st LO
~2000 MHz

TX_PA_EN

J107

Fref_IN
LD1
LO_CLK
LE1
MUX1
RXEN
I2C_SDA
Mute_LO1
LO_DATA
CE1
I2C_SCL

J108

Fref_IN
LD2
LO_CLK
LE2
MUX2
RXEN
I2C_SDA
Mute_LO2
LO_DATA
CE2
LO2_Buff_EN
I2C_SCL

TP101 GND
TP102 GND
TP103 GND
TP104 GND

+3V6 +5V5
J112
LO1 Module PWR

+3V6 +5V5
J113
LO1 Module PWR

+5V5
+3V6

Fref IN

J104 SMA
R101 DNP/0
Fref_IN

DNP when using on board VCTCXOs
0 + DNP LO modules jumpers for VCTCXOs

J105
DIN 41612 09231486921

3V6
IMON
5V5_FLT
MUX2
LO_DATA
3V6
VIN_Sense
PSU_FAULT
Vfwd
ATT_CLK
LD2
RXEN
Mute_LO1
5V5
UART RX
ATT_DATA
MUX1
LD1

3V6_Sense
PCF_INT
ATT_LE
RX_LNA_EN
Mute_LO2
LE1
5V5_Sense
LNBV_Sense
IF_LNA_EN
LO2_Buff_EN
I2C_SCL
LO_CLK
UART_TX
DET_OUT
LE2
I2C_SDA

LO1 Module connector

LO2 Module connector

Control head connector

Digital
File: Digital_IO.kicad_sch

TEMP_D
TX_PA_EN
LNB_FLT
TX_EN
I2C_SCL
I2C_SDA
PCF_INT
UART RX
UART_TX
EXT_UART_RX
INPUT_1
OUTPUT_1
OUTPUT_2
EXT_UART_TX
LNB_POL
LNB_SHDN

IO Port

J111
Conn_01x06

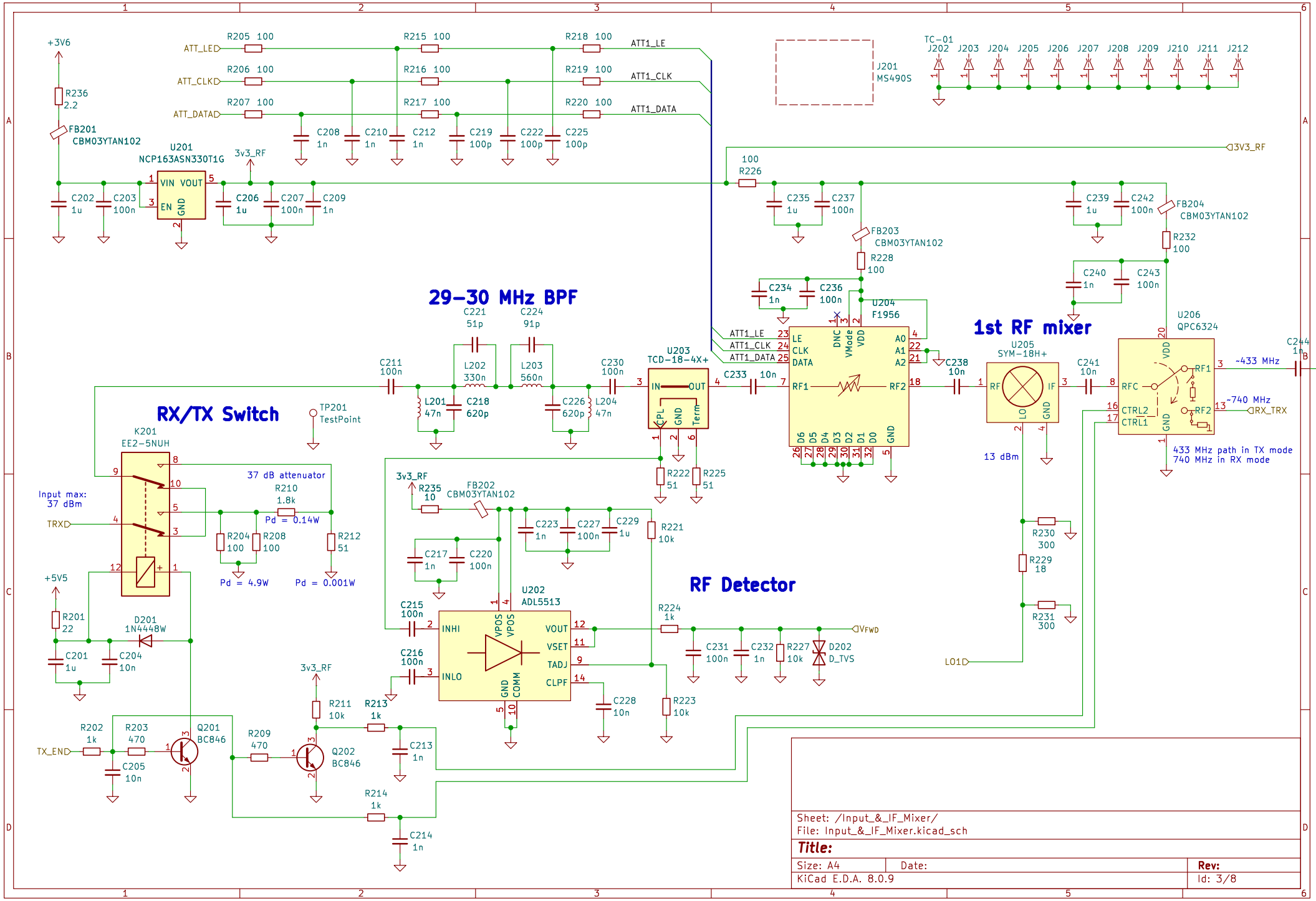
LNB_POL
LNB_SHDN

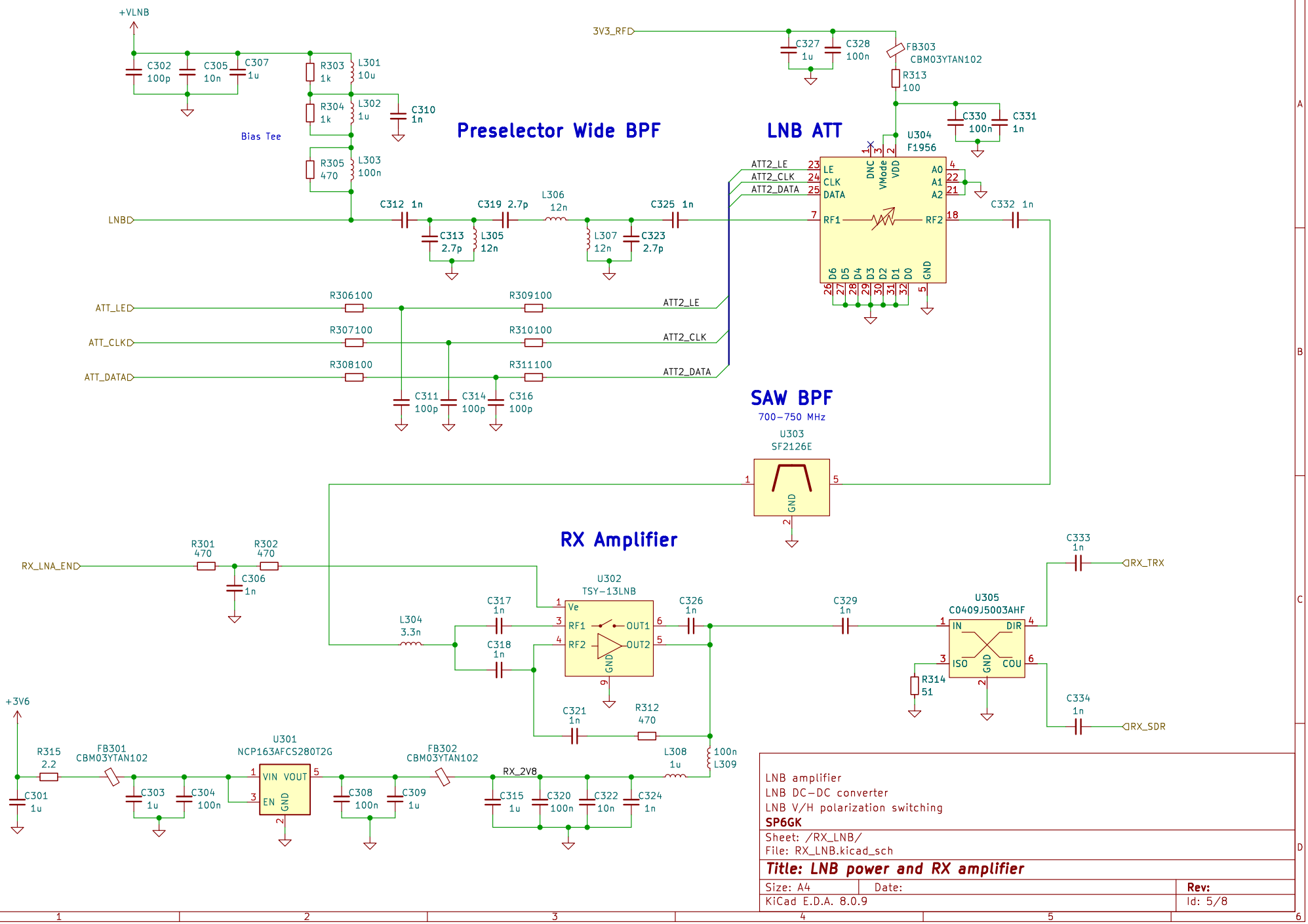
SP6GK

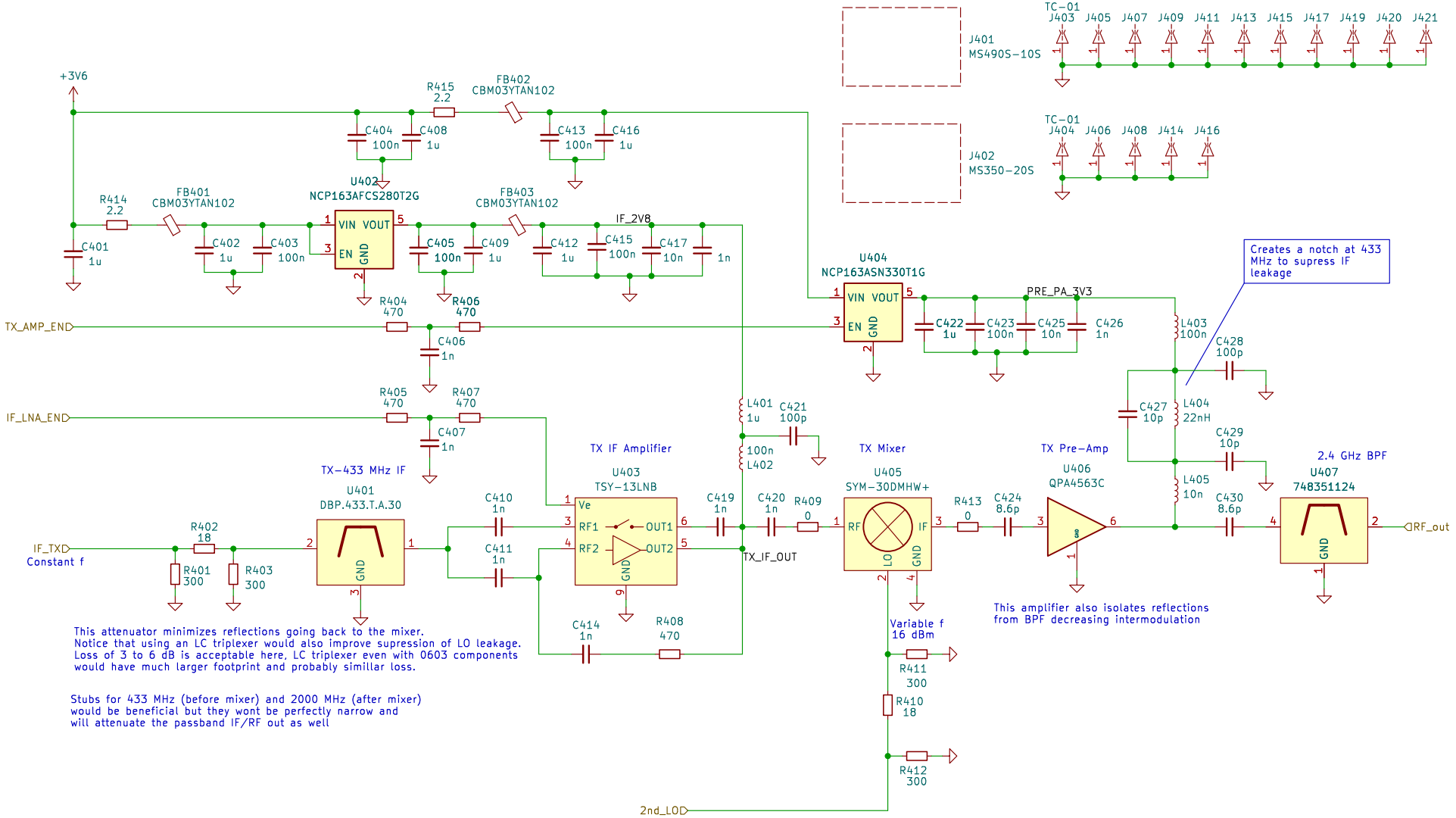
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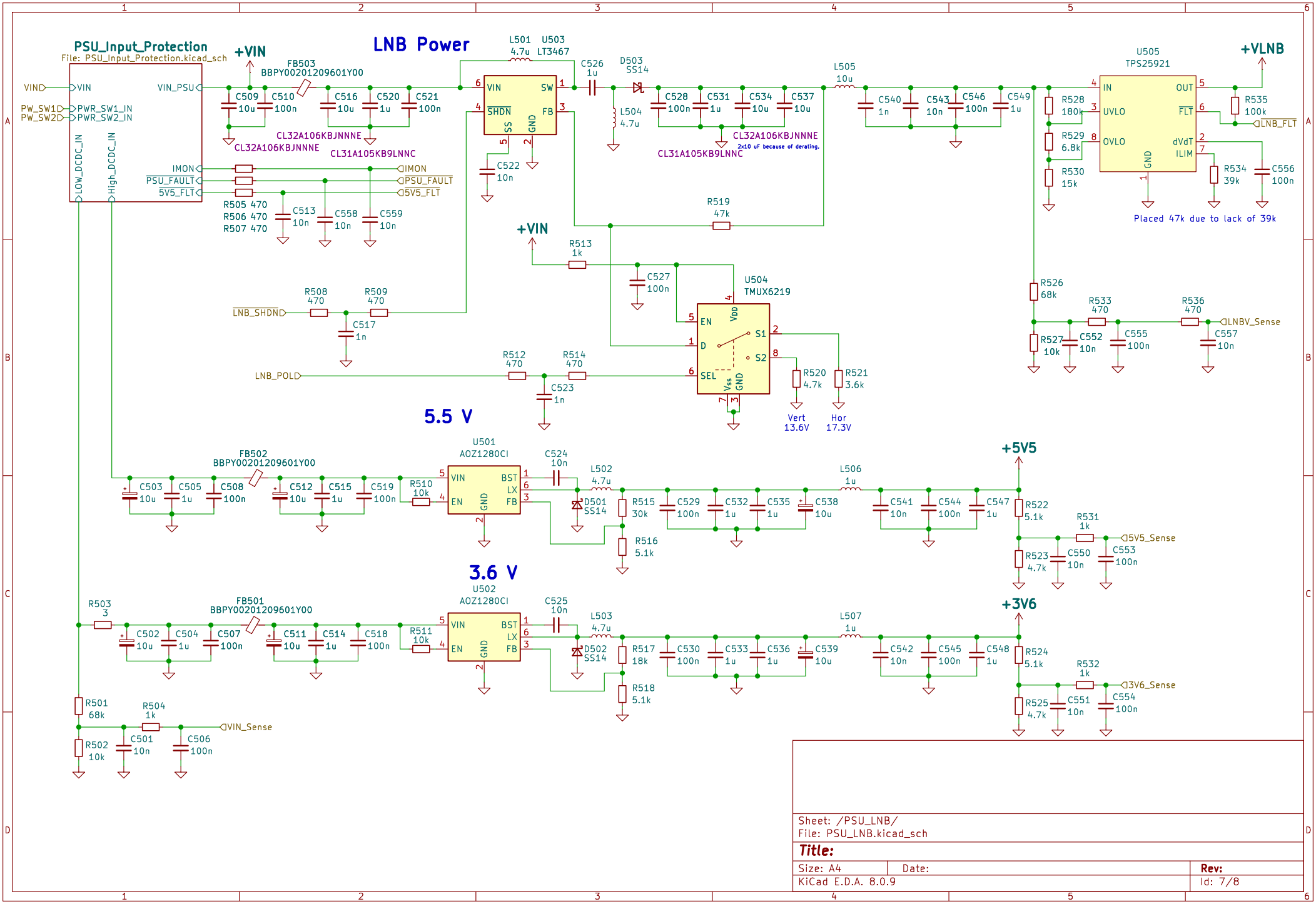
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KiCad E.D.A. 8.0.9		Id: 1/8

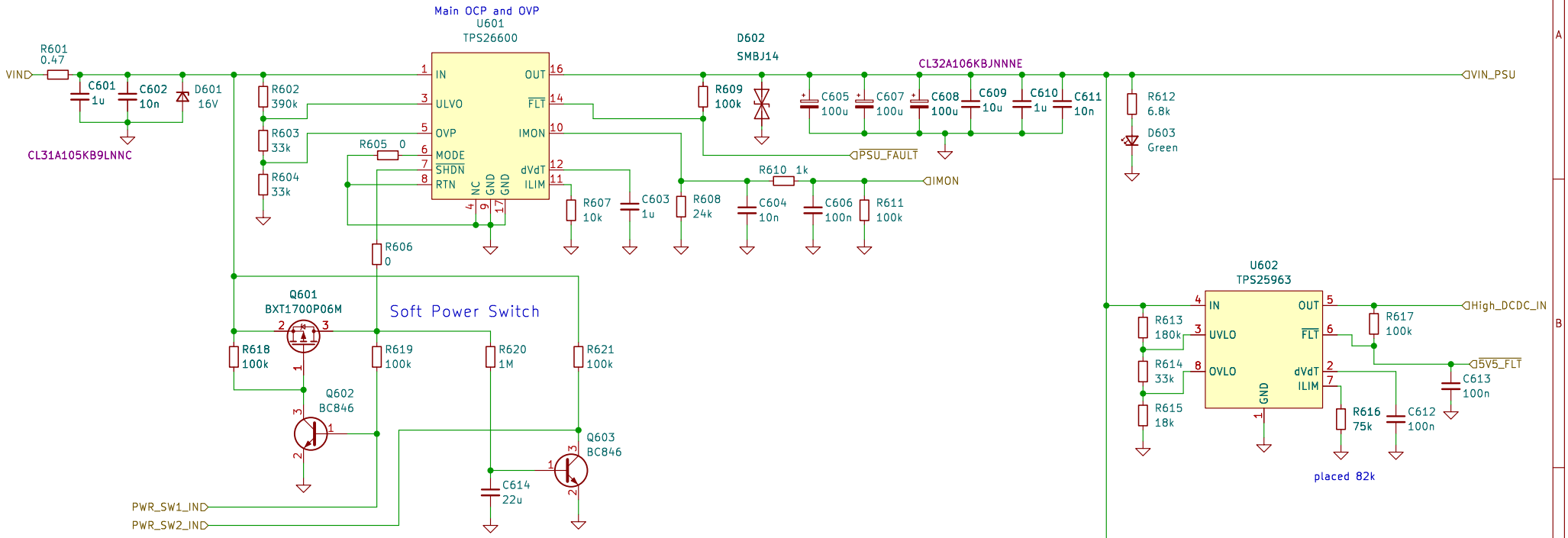

29–30 MHz BPF
RX/TX Switch
RF Detector
1st RF mixer
+3V6
R236 2.2
FB201 CBM03YTAN102
U201 NCP163ASN330T1G
VIN VOUT EN GND
3v3_RF
C202 1u
C203 100n
C206 1u
C207 100n
C209 1n
ATT_LE
ATT_CLK
ATT_DATA
R205 100
R206 100
R207 100
R215 100
R216 100
R217 100
R218 100
R219 100
R220 100
C208 1n
C210 1n
C212 1n
C219 100p
C222 100p
C225 100p
J201 MS490S
TC-01
J202 J203 J204 J205 J206 J207 J208 J209 J210 J211 J212
3V3_RF
R226 100
C235 1u
C237 100n
FB203 CBM03YTAN102
R228 100
C234 1n
C236 100n
U204 F1956
LE CLK DATA DNC VMode VDD
A0 A1 A2
RF1 RF2 GND
C239 1u
C242 100n
FB204 CBM03YTAN102
R232 100
C240 1n
C243 100n
U206 QPC6324
VDD RF1 RFC RF2 CTRL2 CTRL1 GND
~433 MHz
~740 MHz
RX_TRX
433 MHz path in TX mode
740 MHz in RX mode
C244 1n
C221 51p
C224 91p
L202 330n
L203 560n
C211 100n
C230 100n
U203 TCD-18-4X+
IN OUT CPL GND Term
L201 47n
C218 620p
C226 620p
L204 47n
C233 10n
C238 10n
U205 SYM-18H+
RF IF LO GND
13 dBm
C241 10n
TP201 TestPoint
K201 EE2-5NUH
Input max: 37 dBm
TRXD
37 dB attenuator
R210 1.8k
Pd = 0.14W
R204 100
R208 100
R212 51
Pd = 4.9W
Pd = 0.001W
R222 51
R225 51
3v3_RF
R235 10
FB202 CBM03YTAN102
C223 1n
C227 100n
C229 1u
R221 10k
C217 1n
C220 100n
U202 ADL5513
INHI INLO VPOS VPOS GND COMM VOUT VSET TADJ CLPF
C215 100n
C216 100n
R224 1k
VFWD
C231 100n
C232 1n
R227 10k
D202 D_TVS
C228 10n
R223 10k
R230 300
R229 18
R231 300
LO1
+5V5
R201 22
D201 1N4448W
C201 1u
C204 10n
R202 1k
R203 470
Q201 BC846
TX_END
C205 10n
R209 470
Q202 BC846
R211 10k
R213 1k
C213 1n
R214 1k
C214 1n
Sheet: /Input_&_IF_Mixer/
File: Input_&_IF_Mixer.kicad_sch
Title:
Size: A4
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+VLNB
C302 100p
C305 10n
C307 1u
Bias Tee
R303 1k
L301 10u
R304 1k
L302 1u
C310 1n
R305 470
L303 100n
Preselector Wide BPF
3V3_RFD
C327 1u
C328 100n
FB303 CBM03YTAN102
R313 100
C330 100n
C331 1n
LNB ATT
U304 F1956
ATT2_LE
ATT2_CLK
ATT2_DATA
LE
CLK
DATA
DNC
VMode
VDD
A0
A1
A2
RF1
RF2
D6
D5
D4
D3
D2
D1
D0
GND
LNB
C312 1n
C319 2.7p
L306 12n
C325 1n
C313 2.7p
L305 12n
L307 12n
C323 2.7p
C332 1n
ATT_LE
ATT_CLK
ATT_DATA
R306 100
R307 100
R308 100
R309 100
R310 100
R311 100
C311 100p
C314 100p
C316 100p
SAW BPF
700–750 MHz
U303
SF2126E
GND
RX Amplifier
RX_LNA_END
R301 470
R302 470
C306 1n
U302
TSY-13LNB
Ve
RF1
RF2
OUT1
OUT2
GND
L304 3.3n
C317 1n
C318 1n
C326 1n
C329 1n
U305
C0409J5003AHF
IN
DIR
ISO
GND
COU
C333 1n
RX_TRX
R314 51
C334 1n
RX_SDR
C321 1n
R312 470
+3V6
R315 2.2
C301 1u
FB301 CBM03YTAN102
C303 1u
C304 100n
U301
NCP163AFCS280T2G
VIN
VOUT
EN
GND
C308 100n
C309 1u
FB302 CBM03YTAN102
RX_2V8
L308 1u
100n L309
C315 1u
C320 100n
C322 10n
C324 1n
LNB amplifier
LNB DC–DC converter
LNB V/H polarization switching
SP6GK
Sheet: /RX_LNB/
File: RX_LNB.kicad_sch
Title: LNB power and RX amplifier
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.9
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Sheet: /TX_mixer/		
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Title:		
Size: A4	Date:	**Rev:**
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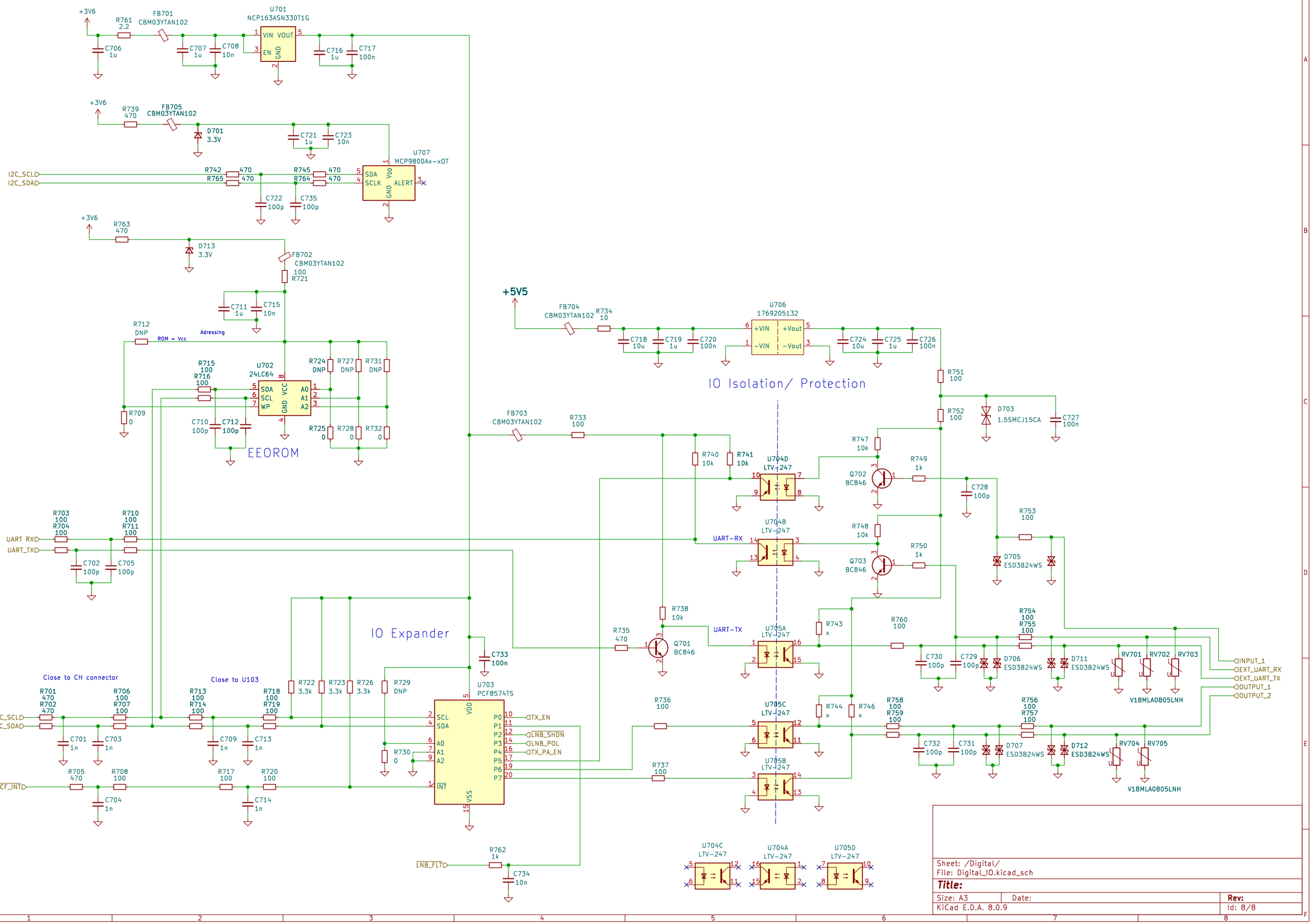

PSU_Input_Protection
File: PSU_Input_Protection.kicad_sch
LNB Power
+VIN
+VLNB
VIN
VIN_PSU
PW_SW1
PW_SW2
PWR_SW1_IN
PWR_SW2_IN
LOW_DCDC_IN
High_DCDC_IN
IMON
PSU_FAULT
5V5_FLT
FB503
BBPY00201209601Y00
L501 4.7u
U503 LT3467
D503 SS14
L504 4.7u
U505 TPS25921
CL32A106KBJNNNE
CL31A105KB9LNNC
2x10 uF because of derating.
Placed 47k due to lack of 39k
LNB_SHDN
LNB_POL
LNB_FLT
U504 TMUX6219
Vert 13.6V
Hor 17.3V
LNBV_Sense
5.5 V
U501 AOZ1280CI
FB502
BBPY00201209601Y00
D501 SS14
+5V5
5V5_Sense
3.6 V
U502 AOZ1280CI
FB501
BBPY00201209601Y00
D502 SS14
+3V6
3V6_Sense
VIN_Sense
Sheet: /PSU_LNB/
File: PSU_LNB.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.9
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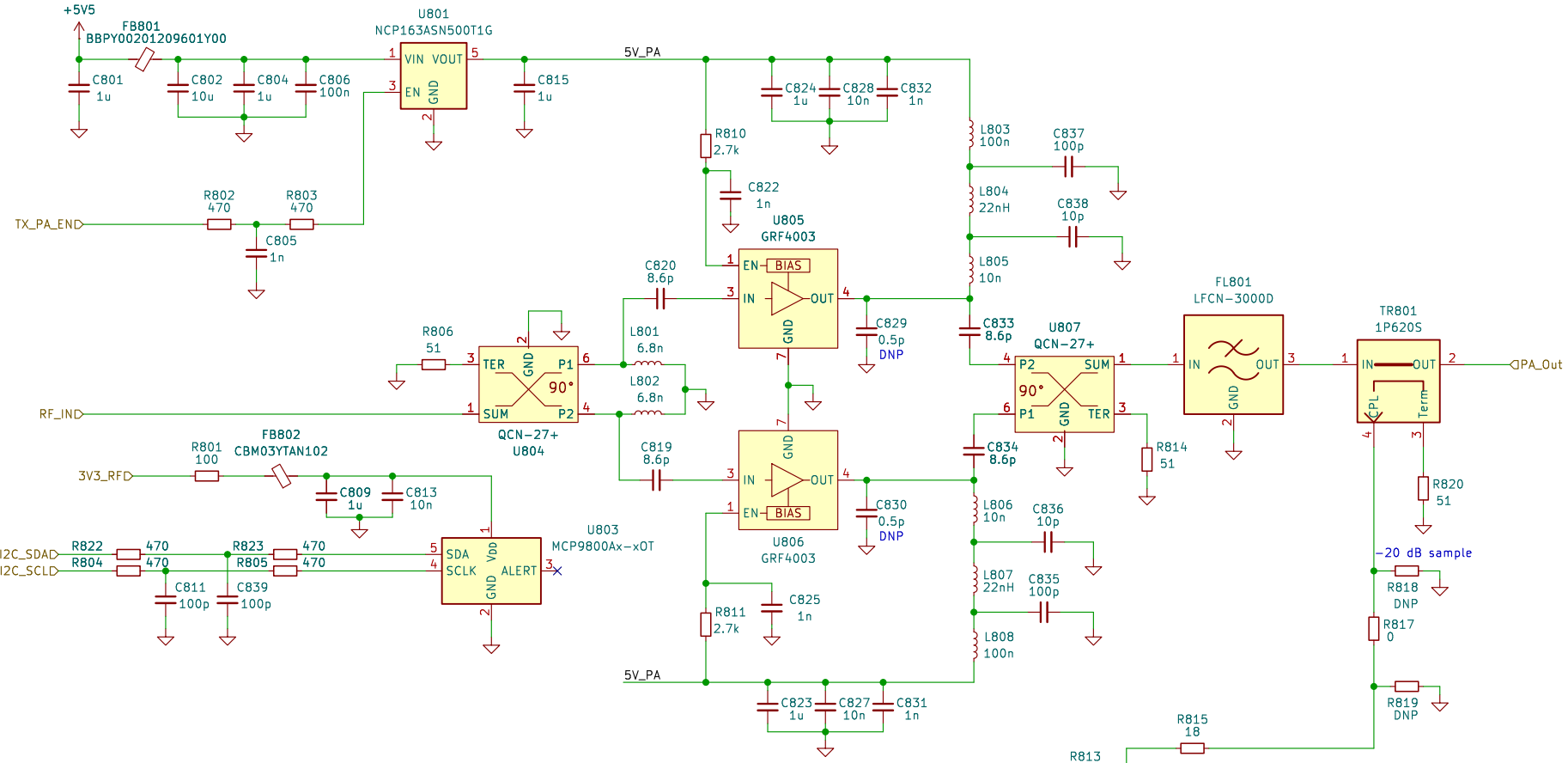


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File: PSU_Input_Protection.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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+3V6
R761 2.2
FB701 CBM03YTAN102
U701 NCP163ASN330T1G
VIN VOUT EN GND
C706 1u
C707 1u
C708 10n
C716 1u
C717 100n
+3V6
R739 470
FB705 CBM03YTAN102
D701 3.3V
C721 1u
C723 10n
U707 MCP9800Ax-xOT
SDA SCLK Vdd ALERT GND
I2C_SCLD
I2C_SDAD
R742 470
R765 470
R745 470
R764 470
C722 100p
C735 100p
+3V6
R763 470
D713 3.3V
FB702 CBM03YTAN102
R721 100
C711 1u
C715 10n
R712 DNP
ROM = Vcc
Adressing
U702 24LC64
SDA SCL WP GND VCC A0 A1 A2
R715 100
R716 100
R724 DNP
R727 DNP
R731 DNP
R709 0
C710 100p
C712 100p
R725 0
R728 0
R732 0
EEOROM
+5V5
FB704 CBM03YTAN102
R734 10
U706 1769205132
+VIN -VIN +Vout -Vout
C718 10u
C719 1u
C720 100n
C724 10u
C725 1u
C726 100n
IO Isolation/ Protection
R751 100
R752 100
D703 1.5SMCJ15CA
C727 100n
FB703 CBM03YTAN102
R733 100
R740 10k
R741 10k
U704D LTV-247
R747 10k
Q702 BC846
R749 1k
C728 100p
R753 100
U704B LTV-247
UART-RX
R748 10k
Q703 BC846
R750 1k
D705 ESD3B24WS
R703 100
R704 100
R710 100
R711 100
UART RX
UART TX
C702 100p
C705 100p
IO Expander
R738 10k
R735 470
Q701 BC846
UART-TX
U705A LTV-247
R743 x
R760 100
R754 100
R755 100
C730 100p
C729 100p
D706 ESD3B24WS
D711 ESD3B24WS
RV701
RV702
RV703
V18MLA0805LNH
INPUT_1
EXT_UART_RX
EXT_UART_TX
OUTPUT_1
OUTPUT_2
C733 100n
Close to CH connector
Close to U103
R722 3.3k
R723 3.3k
R726 3.3k
R729 DNP
U703 PCF8574TS
SCL SDA A0 A1 A2 INT VDD VSS P0 P1 P2 P3 P4 P5 P6 P7
TX_EN
LNB_SHDN
LNB_POL
TX_PA_EN
R701 470
R702 470
R706 100
R707 100
R713 100
R714 100
R718 100
R719 100
I2C_SCLD
I2C_SDAD
C701 1n
C703 1n
C709 1n
C713 1n
R730 0
R736 100
U705C LTV-247
R744 x
R746 x
R758 100
R759 100
R756 100
R757 100
C732 100p
C731 100p
D707 ESD3B24WS
D712 ESD3B24WS
RV704
RV705
V18MLA0805LNH
PCF_INTD
R705 470
R708 100
R717 100
R720 100
C704 1n
C714 1n
R737 100
U705B LTV-247
R762 1k
LNB_FLTD
C734 10n
U704C LTV-247
U704A LTV-247
U705D LTV-247
Sheet: /Digital/
File: Digital_IO.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 8.0.9
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+5V5
FB801
BBPY00201209601Y00
U801
NCP163ASN500T1G
VIN VOUT
EN GND
C801 1u
C802 10u
C804 1u
C806 100n
C815 1u
5V_PA
C824 1u
C828 10n
C832 1n
R810 2.7k
C822 1n
L803 100n
C837 100p
L804 22nH
C838 10p
L805 10n
TX_PA_EN
R802 470
R803 470
C805 1n
U805
GRF4003
EN BIAS
IN OUT
GND
C820 8.6p
C829 0.5p DNP
C833 8.6p
U807
QCN-27+
FL801
LFCN-3000D
TR801
1P620S
IN OUT
GND
CPL Term
PA_Out
R806 51
L801 6.8n
L802 6.8n
TER GND P1
90°
SUM P2
RF_IN
QCN-27+
U804
FB802
CBM03YTAN102
R801 100
3V3_RF
C809 1u
C813 10n
C819 8.6p
C834 8.6p
R814 51
R820 51
L806 10n
C836 10p
U806
GRF4003
C830 0.5p DNP
U803
MCP9800Ax-xOT
SDA VDD
SCLK ALERT
GND
I2C_SDA
I2C_SCL
R822 470
R804 470
R823 470
R805 470
C811 100p
C839 100p
L807 22nH
C835 100p
-20 dB sample
R818 DNP
R817 0
R819 DNP
R811 2.7k
C825 1n
L808 100n
5V_PA
C823 1u
C827 10n
C831 1n
R815 18
R813 18
R816 18
Sample
R821 10
FB803
CBM03YTAN102
3V3_RF
C803 1u
C814 1n
C816 100n
C818 1u
R807 10k
C810 1n
C812 100n
RF Detector
U802
ADL5513
C807 10p
INHI
VPOS VPOS
VOUT
VSET
TADJ
CLPF
INLO
GND COMM
C808 100n
C817 10n
R809 1k
DET_OUT
C821 100n
C826 1n
R812 10k
D801
D_TVS
R808 10k
Sheet: /Output_PA/
File: Output_PA.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.9
Id: 8/8